

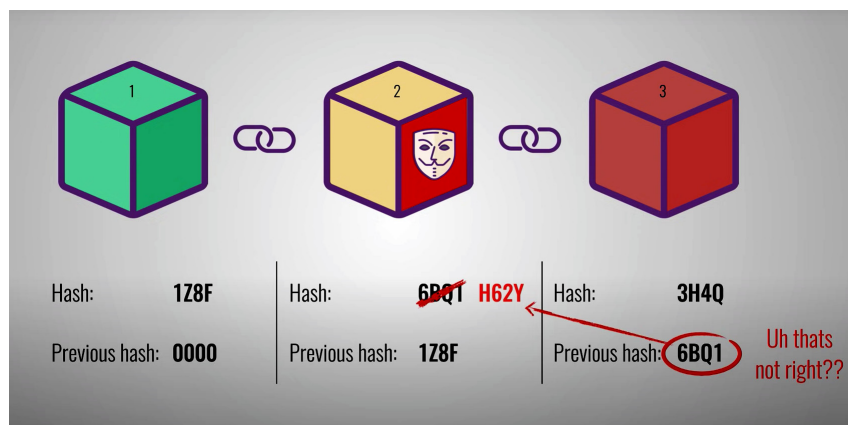


Figure 4: Tampering with data in a blockchain (Courtesy: Simply Explained)

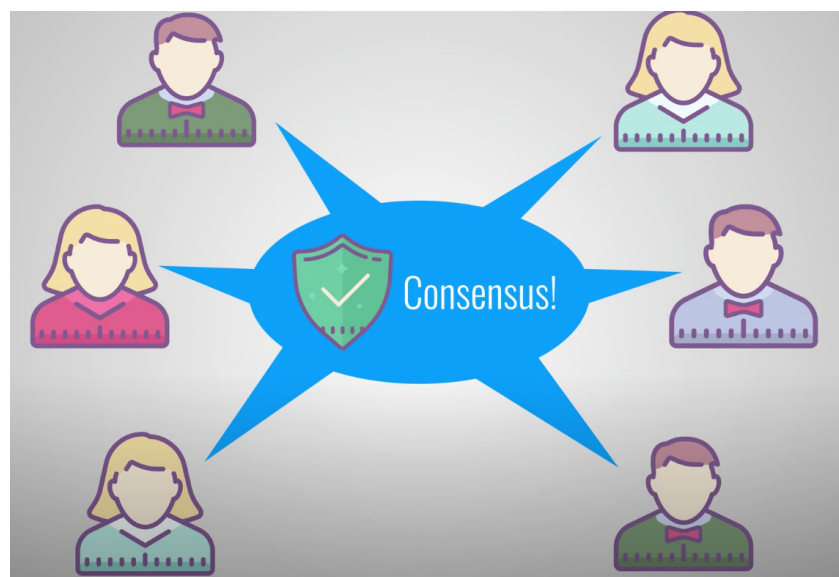


Figure 5: Decentralisation in blockchain (Courtesy: Simply Explained)

The decentralisation of blockchain technology is widely appreciated. But what exactly does that imply? How are blockchains able to function without a central authority? This is accomplished through the use of a 'distributed ledger'. When you are a part of a blockchain, such as Bitcoin, what you are, in reality, is a node in a vast network of other nodes engaged in monetary transactions. Instead of having a single node control and maintaining the primary ledger, blockchain distributes a copy of the ledger to all nodes in the network. So, if you have Bitcoin software installed on your computer, you are a node in

the Bitcoin network and your computer has a ledger of the entire blockchain. As shown in Figure 5, any time a new block is added to the chain, all nodes in the network are notified of the change, and are required to verify and accept the new block as a legal transaction, building a consensus among all parties involved. Because no single node has complete control over the chain, blockchain is referred to as a decentralised network. Instead, each node has equal power, and all modifications are reflected in the ledgers of all nodes in the network, ensuring that all transactions are completely transparent and secure.

What is open source software and why does it matter?

Simply put, open source software is software for which the source code is publicly available. The term 'open source' is used primarily in conjunction with software, but it is a methodology that can be applied to anything. In fact, open source is used for collaborative development of many different technologies apart from software, such as Web development frameworks, high/low level computer languages, bots, plugins and, most importantly, blockchains.

From a distance, the incentive for creating open source software is similar to the rationale for creating a blockchain. The founders of the open source movement aimed to make the software development process more transparent. If a software's source code is made public, essential information such as security issues, the type of user data collected and saved by the software, and the many frameworks and programming languages used to create the software are all made public. This grants the engineers the freedom to read and understand the program at a granular level, and to contribute or modify it to meet their own needs and objectives.

Linus Torvalds' operating system, Linux, rose to prominence in the late 1990s. Linux is now found in everything — from cell phones to televisions and the Mars Rover. It was, nevertheless, a simple yet groundbreaking piece of technology unlike any other at the time. Its source code was open source and freely available. Publishing your source code openly was unthinkable to many in the computer business at the time, but the benefits began to outweigh the drawbacks, and soon enough, more and more organisations joined the bandwagon. Netscape was the first significant name in the business to open up the source code of its Web browser. Soon after, in 1998, a meeting was convened in